Clustering, $m = 5$ labelled windows per cluster

Table 1: Clustering, L1 (per-house activities, resident id removed), rare classes masked; $m = 5$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (100 labels)									
k-means	Sensor hist.	0.70	0.46	0.88	0.65	0.25	0.26	0.61	0.45
k-means	Sensor+ToD hist.	0.70	0.37	0.77	0.34	0.78	0.50	0.75	0.41
k-means	MLM	0.77	0.50	**0.89**	**0.66**	0.30	0.27	0.65	0.48
k-means	MLM-time	0.72	0.41	0.88	0.66	0.30	0.29	0.63	0.45
k-means	MLM + ToD	0.71	0.32	0.80	0.35	0.79	**0.53**	0.77	0.40
k-means	MLM-time + ToD	0.75	0.40	0.76	0.39	**0.81**	0.52	**0.77**	0.44
SCAN	MLM	0.76	0.40	0.84	0.48	0.38	0.34	0.66	0.41
SCAN	MLM-time	0.75	0.49	0.88	0.62	0.33	0.29	0.65	0.47
DEC	MLM	0.77	0.46	0.88	0.61	0.23	0.27	0.63	0.45
DEC	MLM-time	**0.78**	**0.52**	**0.89**	0.62	0.30	0.30	0.66	**0.48**
$K = 50$ clusters (250 labels)									
k-means	Sensor hist.	0.78	0.57	**0.89**	**0.66**	0.33	0.32	0.67	0.52
k-means	Sensor+ToD hist.	0.77	0.50	0.86	0.61	0.80	**0.62**	0.81	0.58
k-means	MLM	0.79	**0.61**	**0.89**	0.65	0.37	0.35	0.68	0.54
k-means	MLM-time	0.80	0.60	**0.88**	0.65	0.34	0.29	0.68	0.52
k-means	MLM + ToD	0.79	0.52	0.87	0.58	0.80	0.59	0.82	0.56
k-means	MLM-time + ToD	**0.81**	0.57	0.84	0.60	**0.82**	0.58	**0.83**	**0.59**
SCAN	MLM	0.79	0.55	0.86	0.51	0.36	0.36	0.67	0.48
SCAN	MLM-time	0.78	0.60	0.85	0.62	0.35	0.34	0.66	0.52
DEC	MLM	0.78	0.54	0.88	0.61	0.36	0.33	0.67	0.49
DEC	MLM-time	0.80	0.57	**0.88**	0.63	0.31	0.29	0.66	0.50
$K = 100$ clusters (500 labels)									
k-means	Sensor hist.	0.78	0.52	**0.89**	**0.66**	0.34	0.34	0.67	0.51
k-means	Sensor+ToD hist.	**0.80**	0.57	0.87	0.63	0.81	0.61	0.83	0.60
k-means	MLM	0.79	0.57	**0.89**	0.65	0.39	0.39	0.69	0.53
k-means	MLM-time	0.79	**0.60**	**0.88**	0.65	0.34	0.34	0.67	0.53
k-means	MLM + ToD	**0.80**	0.55	**0.88**	0.63	**0.83**	**0.65**	**0.84**	**0.61**
k-means	MLM-time + ToD	0.80	0.54	0.87	0.62	**0.83**	0.64	**0.83**	0.60
SCAN	MLM	**0.80**	0.58	0.87	0.58	0.38	0.39	0.68	0.52
SCAN	MLM-time	0.79	0.53	0.86	0.56	0.36	0.37	0.67	0.49
DEC	MLM	0.76	0.51	0.88	0.61	0.33	0.33	0.66	0.48
DEC	MLM-time	0.78	0.53	**0.89**	0.63	0.35	0.35	0.67	0.50

Table 2: Clustering, L1, all classes; $m = 5$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (100 labels)									
k-means	Sensor hist.	0.69	0.35	0.88	0.46	0.25	0.27	0.61	0.36
k-means	Sensor+ToD hist.	0.69	0.27	0.77	0.24	0.77	0.46	0.75	0.32
k-means	MLM	0.76	0.37	**0.89**	**0.47**	0.30	0.24	0.65	0.36
k-means	MLM-time	0.72	0.30	0.88	0.46	0.30	0.27	0.63	0.34
k-means	MLM + ToD	0.71	0.24	0.79	0.25	0.79	**0.48**	0.76	0.32
k-means	MLM-time + ToD	0.74	0.30	0.76	0.27	**0.81**	0.48	**0.77**	0.35
SCAN	MLM	0.75	0.29	0.83	0.34	0.38	0.31	0.65	0.31
SCAN	MLM-time	0.74	0.36	0.87	0.44	0.33	0.27	0.65	0.35
DEC	MLM	0.76	0.34	0.88	0.42	0.23	0.25	0.62	0.34
DEC	MLM-time	**0.77**	**0.38**	**0.89**	0.43	0.30	0.27	0.65	**0.36**
$K = 50$ clusters (250 labels)									
k-means	Sensor hist.	0.78	0.42	**0.89**	0.46	0.33	0.32	0.66	0.40
k-means	Sensor+ToD hist.	0.77	0.37	0.86	0.43	0.80	**0.56**	0.81	0.45
k-means	MLM	0.79	**0.45**	**0.88**	0.45	0.37	0.36	0.68	0.42
k-means	MLM-time	0.80	0.44	**0.88**	0.46	0.34	0.27	0.67	0.39
k-means	MLM + ToD	0.79	0.38	0.87	0.41	0.80	0.55	0.82	0.45
k-means	MLM-time + ToD	**0.81**	0.42	0.84	0.42	**0.82**	0.53	**0.82**	**0.46**
SCAN	MLM	0.78	0.44	0.85	0.45	0.36	0.36	0.66	0.41
SCAN	MLM-time	0.78	**0.45**	0.84	**0.51**	0.35	0.31	0.66	0.42
DEC	MLM	0.78	0.42	0.88	0.43	0.36	0.30	0.67	0.38
DEC	MLM-time	0.79	0.42	**0.88**	0.44	0.30	0.26	0.66	0.37
$K = 100$ clusters (500 labels)									
k-means	Sensor hist.	0.77	0.45	**0.88**	0.46	0.34	0.34	0.67	0.42
k-means	Sensor+ToD hist.	**0.80**	0.43	0.87	0.44	0.80	**0.60**	0.82	**0.49**
k-means	MLM	0.78	0.47	**0.89**	0.45	0.39	0.40	0.69	0.44
k-means	MLM-time	0.79	0.44	**0.88**	0.45	0.34	0.31	0.67	0.40
k-means	MLM + ToD	**0.79**	0.41	**0.88**	0.44	**0.82**	0.59	**0.83**	0.48
k-means	MLM-time + ToD	0.79	0.40	0.87	0.45	**0.82**	0.59	**0.83**	0.48
SCAN	MLM	**0.80**	**0.48**	0.87	**0.52**	0.38	0.35	0.68	0.45
SCAN	MLM-time	0.79	0.39	0.85	0.47	0.36	0.38	0.67	0.42
DEC	MLM	0.76	0.41	0.88	0.43	0.33	0.30	0.66	0.38
DEC	MLM-time	0.78	0.44	**0.88**	0.47	0.35	0.32	0.67	0.41

Table 3: Clustering, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS; $m = 5$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (100 labels)									
k-means	Sensor hist.	0.70	0.38	0.88	0.50	0.73	0.29	0.77	0.39
k-means	Sensor+ToD hist.	0.71	0.28	0.78	0.32	0.84	0.39	0.78	0.33
k-means	MLM	0.77	0.33	**0.89**	**0.51**	0.74	0.29	0.80	0.38
k-means	MLM-time	0.73	0.32	0.89	0.50	0.77	0.32	0.79	0.38
k-means	MLM + ToD	0.72	0.29	0.81	0.32	**0.86**	**0.44**	0.80	0.35
k-means	MLM-time + ToD	0.74	0.32	0.76	0.31	**0.86**	0.41	0.79	0.35
SCAN	MLM	0.76	0.33	0.85	0.36	0.77	0.33	0.79	0.34
SCAN	MLM-time	0.75	0.39	0.88	0.49	0.77	0.34	0.80	**0.41**
DEC	MLM	**0.78**	0.38	0.89	0.47	0.77	0.32	**0.81**	0.39
DEC	MLM-time	**0.78**	**0.40**	**0.89**	0.47	0.78	0.35	**0.82**	**0.41**
$K = 50$ clusters (250 labels)									
k-means	Sensor hist.	0.78	0.43	**0.89**	0.50	0.75	0.37	0.81	0.43
k-means	Sensor+ToD hist.	0.77	0.40	0.85	0.46	0.87	0.43	0.83	0.43
k-means	MLM	**0.80**	**0.46**	**0.89**	0.50	0.78	0.40	0.82	0.45
k-means	MLM-time	**0.80**	**0.46**	**0.89**	0.50	0.76	0.29	0.82	0.42
k-means	MLM + ToD	0.79	0.41	0.88	0.47	0.86	**0.49**	**0.85**	0.46
k-means	MLM-time + ToD	**0.80**	0.42	0.85	0.47	**0.87**	0.43	**0.84**	0.44
SCAN	MLM	0.79	0.45	0.85	0.53	0.79	0.42	0.81	**0.47**
SCAN	MLM-time	0.78	**0.46**	0.85	**0.56**	0.78	0.36	0.80	0.46
DEC	MLM	0.79	0.43	**0.88**	0.47	0.79	0.35	0.82	0.42
DEC	MLM-time	0.80	0.43	**0.89**	0.47	0.76	0.32	0.82	0.41
$K = 100$ clusters (500 labels)									
k-means	Sensor hist.	0.78	0.38	**0.89**	0.50	0.77	0.39	0.81	0.42
k-means	Sensor+ToD hist.	0.79	0.43	0.87	0.47	**0.87**	**0.50**	0.85	0.47
k-means	MLM	0.80	0.46	**0.89**	0.49	0.80	0.44	0.83	0.46
k-means	MLM-time	0.79	0.46	0.89	0.49	0.79	0.35	0.82	0.43
k-means	MLM + ToD	**0.81**	0.43	0.89	0.48	**0.88**	0.48	**0.86**	0.46
k-means	MLM-time + ToD	0.80	0.43	0.88	0.51	**0.87**	0.48	0.85	0.48
SCAN	MLM	0.80	**0.47**	0.88	**0.58**	0.81	0.41	0.83	**0.48**
SCAN	MLM-time	0.79	0.41	0.85	0.55	0.81	0.46	0.82	0.47
DEC	MLM	0.78	0.40	0.88	0.46	0.79	0.35	0.82	0.41
DEC	MLM-time	0.79	0.41	**0.89**	0.51	0.79	0.35	0.82	0.42

Clustering, $m = 1$ labelled windows per cluster

Table 4: Clustering, L1 (per-house activities, resident id removed), rare classes masked; $m = 1$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (20 labels)									
k-means	Sensor hist.	0.70	0.46	0.82	0.63	0.21	0.24	0.58	0.45
k-means	Sensor+ToD hist.	0.69	0.36	0.76	0.35	0.77	0.48	**0.74**	0.40
k-means	MLM	0.73	0.38	**0.89**	**0.66**	0.25	0.24	0.62	0.43
k-means	MLM-time	0.73	0.41	0.84	0.55	0.32	0.20	0.63	0.39
k-means	MLM + ToD	0.62	0.29	0.79	0.35	**0.79**	**0.53**	0.74	0.39
k-means	MLM-time + ToD	0.72	0.34	0.71	0.38	0.77	0.50	0.73	0.41
SCAN	MLM	**0.76**	0.43	0.77	0.47	0.35	0.31	0.63	0.40
SCAN	MLM-time	0.75	**0.52**	0.82	0.60	0.33	0.29	0.63	0.47
DEC	MLM	**0.76**	0.47	0.88	0.61	0.23	0.28	0.62	0.45
DEC	MLM-time	**0.76**	0.49	0.86	0.62	0.37	0.33	0.66	**0.48**
$K = 50$ clusters (50 labels)									
k-means	Sensor hist.	0.70	0.42	0.84	**0.64**	0.31	0.30	0.62	0.45
k-means	Sensor+ToD hist.	0.71	0.50	0.81	0.55	0.79	0.52	0.77	0.52
k-means	MLM	0.77	**0.58**	0.86	0.62	0.36	0.36	0.66	0.52
k-means	MLM-time	**0.77**	0.49	0.84	0.61	0.37	0.34	0.66	0.48
k-means	MLM + ToD	0.72	0.47	0.85	0.51	0.79	**0.55**	0.79	0.51
k-means	MLM-time + ToD	**0.77**	0.50	0.82	0.60	**0.80**	0.52	**0.80**	**0.54**
SCAN	MLM	0.76	0.51	0.83	0.42	0.31	0.29	0.63	0.41
SCAN	MLM-time	0.74	0.51	0.85	0.61	0.37	0.32	0.65	0.48
DEC	MLM	0.74	0.50	0.86	0.60	0.35	0.31	0.65	0.47
DEC	MLM-time	**0.77**	0.53	**0.87**	**0.63**	0.35	0.31	0.66	0.49
$K = 100$ clusters (100 labels)									
k-means	Sensor hist.	0.74	0.55	**0.87**	0.63	0.31	0.29	0.64	0.49
k-means	Sensor+ToD hist.	0.70	0.50	0.86	0.62	**0.80**	0.57	0.79	**0.56**
k-means	MLM	0.76	**0.59**	0.85	**0.64**	0.39	0.37	0.67	0.53
k-means	MLM-time	0.73	0.53	0.86	0.63	0.34	0.32	0.64	0.49
k-means	MLM + ToD	**0.78**	0.53	0.82	0.59	**0.80**	**0.58**	**0.80**	**0.57**
k-means	MLM-time + ToD	0.75	0.54	0.84	0.62	0.78	0.51	0.79	0.56
SCAN	MLM	0.74	0.51	0.85	0.56	0.34	0.34	0.64	0.47
SCAN	MLM-time	0.76	0.50	0.83	0.53	0.36	0.36	0.65	0.47
DEC	MLM	0.74	0.47	**0.87**	0.60	0.32	0.30	0.65	0.46
DEC	MLM-time	0.75	0.49	**0.87**	0.62	0.32	0.32	0.65	0.48

Table 5: Clustering, L1, all classes; $m = 1$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ *clusters (20 labels)*									
k-means	Sensor hist.	0.69	0.34	0.82	0.44	0.21	0.26	0.58	0.35
k-means	Sensor+ToD hist.	0.69	0.27	0.76	0.24	0.77	0.43	**0.74**	0.31
k-means	MLM	0.73	0.28	**0.89**	**0.46**	0.25	0.22	0.62	0.32
k-means	MLM-time	0.72	0.30	0.84	0.38	0.32	0.18	0.63	0.29
k-means	MLM + ToD	0.62	0.21	0.79	0.25	**0.79**	**0.48**	0.73	0.31
k-means	MLM-time + ToD	0.72	0.25	0.71	0.27	0.76	0.46	0.73	0.32
SCAN	MLM	**0.75**	0.31	0.77	0.33	0.35	0.28	0.62	0.31
SCAN	MLM-time	0.75	**0.38**	0.82	0.42	0.32	0.27	0.63	0.36
DEC	MLM	**0.75**	0.34	0.87	0.42	0.23	0.25	0.62	0.34
DEC	MLM-time	**0.76**	0.36	0.86	0.44	0.36	0.30	0.66	**0.36**
$K = 50$ *clusters (50 labels)*									
k-means	Sensor hist.	0.69	0.31	0.84	0.45	0.31	0.30	0.61	0.35
k-means	Sensor+ToD hist.	0.70	0.37	0.81	0.38	0.79	0.48	0.77	0.41
k-means	MLM	0.76	**0.47**	0.86	0.43	0.36	0.36	0.66	**0.42**
k-means	MLM-time	**0.77**	0.36	0.84	0.43	0.36	0.31	0.66	0.37
k-means	MLM + ToD	0.71	0.36	0.85	0.36	0.78	**0.50**	0.78	0.41
k-means	MLM-time + ToD	**0.76**	0.36	0.82	0.42	**0.80**	0.48	**0.79**	**0.42**
SCAN	MLM	0.75	0.38	0.82	0.30	0.31	0.29	0.63	0.32
SCAN	MLM-time	0.74	0.38	0.84	**0.50**	0.37	0.29	0.65	0.39
DEC	MLM	0.74	0.36	0.86	0.42	0.35	0.28	0.65	0.36
DEC	MLM-time	**0.77**	0.39	**0.87**	0.44	0.35	0.28	0.66	0.37
$K = 100$ *clusters (100 labels)*									
k-means	Sensor hist.	0.73	0.41	**0.87**	0.44	0.31	0.30	0.64	0.38
k-means	Sensor+ToD hist.	0.70	0.37	0.86	0.43	**0.80**	**0.55**	0.78	**0.45**
k-means	MLM	0.76	**0.49**	0.85	**0.45**	0.38	0.36	0.67	0.43
k-means	MLM-time	0.73	0.39	0.86	**0.44**	0.34	0.29	0.64	0.37
k-means	MLM + ToD	**0.78**	0.39	0.82	0.41	**0.80**	0.53	**0.80**	0.44
k-means	MLM-time + ToD	0.75	0.40	0.84	0.44	0.78	0.46	0.79	0.43
SCAN	MLM	0.74	0.43	0.85	0.39	0.34	0.31	0.64	0.38
SCAN	MLM-time	0.76	0.38	0.83	0.37	0.36	0.35	0.65	0.37
DEC	MLM	0.74	0.35	**0.87**	0.42	0.32	0.28	0.64	0.35
DEC	MLM-time	0.74	0.36	**0.87**	0.43	0.32	0.29	0.64	0.36

Table 6: Clustering, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS; $m = 1$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (20 labels)									
k-means	Sensor hist.	0.71	0.37	0.81	0.45	0.70	0.35	0.74	0.39
k-means	Sensor+ToD hist.	0.69	0.28	0.76	0.29	0.81	0.36	0.76	0.31
k-means	MLM	0.74	0.32	**0.89**	**0.51**	0.72	0.27	0.78	0.37
k-means	MLM-time	0.74	0.32	0.84	0.41	0.74	0.26	0.78	0.33
k-means	MLM + ToD	0.63	0.26	0.80	0.32	**0.86**	**0.44**	0.76	0.34
k-means	MLM-time + ToD	0.73	0.29	0.71	0.29	0.82	0.42	0.75	0.33
SCAN	MLM	0.76	0.33	0.77	0.32	0.78	0.35	0.77	0.34
SCAN	MLM-time	0.75	**0.39**	0.81	0.46	0.78	0.36	0.78	**0.40**
DEC	MLM	0.76	0.36	0.88	0.46	0.73	0.31	0.79	0.38
DEC	MLM-time	**0.76**	0.36	0.86	0.46	0.78	0.34	**0.80**	0.39
$K = 50$ clusters (50 labels)									
k-means	Sensor hist.	0.70	0.32	0.83	0.47	0.74	0.35	0.76	0.38
k-means	Sensor+ToD hist.	0.72	0.41	0.80	0.40	0.83	**0.43**	0.78	0.41
k-means	MLM	**0.78**	**0.46**	0.86	0.46	0.75	0.40	0.80	**0.44**
k-means	MLM-time	0.77	0.36	0.84	0.45	0.80	0.37	0.80	0.39
k-means	MLM + ToD	0.73	0.39	0.87	0.45	0.82	0.41	0.80	0.42
k-means	MLM-time + ToD	0.77	0.40	0.82	0.45	**0.86**	0.40	**0.82**	0.42
SCAN	MLM	0.76	0.42	0.83	0.35	0.71	0.33	0.77	0.37
SCAN	MLM-time	0.75	0.42	0.84	**0.54**	0.79	0.35	0.79	0.44
DEC	MLM	0.75	0.39	0.86	0.46	0.77	0.33	0.79	0.40
DEC	MLM-time	**0.77**	0.39	**0.87**	0.47	0.75	0.31	0.80	0.39
$K = 100$ clusters (100 labels)									
k-means	Sensor hist.	0.74	0.42	**0.88**	**0.47**	0.72	0.35	0.78	0.42
k-means	Sensor+ToD hist.	0.71	0.40	0.86	0.47	0.84	**0.46**	0.80	0.44
k-means	MLM	0.77	**0.46**	0.85	**0.48**	0.77	0.40	0.80	0.45
k-means	MLM-time	0.74	0.41	0.85	0.47	0.74	0.31	0.78	0.40
k-means	MLM + ToD	**0.79**	0.41	0.82	0.44	**0.86**	**0.46**	**0.82**	0.44
k-means	MLM-time + ToD	0.76	0.44	0.84	0.46	**0.86**	**0.46**	**0.82**	**0.46**
SCAN	MLM	0.74	0.40	0.85	0.43	0.73	0.35	0.78	0.39
SCAN	MLM-time	0.76	0.39	0.83	0.43	0.78	0.41	0.79	0.41
DEC	MLM	0.75	0.37	**0.87**	0.45	0.74	0.32	0.79	0.38
DEC	MLM-time	0.76	0.37	**0.87**	0.46	0.76	0.34	0.80	0.39

Clustering, $m = all$ labelled windows per cluster

Table 7: Clustering, L1 (per-house activities, resident id removed), rare classes masked; $m = all$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ *clusters*									
k-means	Sensor hist.	0.79	0.58	0.87	0.60	0.32	0.32	0.66	0.50
k-means	Sensor+ToD hist.	0.70	0.38	0.80	0.38	0.78	0.51	0.76	0.42
k-means	MLM	0.80	**0.59**	**0.89**	**0.65**	0.34	0.32	0.67	0.52
k-means	MLM-time	**0.80**	**0.59**	**0.88**	**0.65**	0.35	0.35	0.68	**0.53**
k-means	MLM + ToD	0.73	0.37	0.80	0.37	0.78	**0.52**	**0.77**	0.42
k-means	MLM-time + ToD	0.72	0.37	0.78	0.32	**0.79**	0.51	0.76	0.40
SCAN	MLM	0.77	0.47	0.84	0.46	0.36	0.33	0.66	0.42
SCAN	MLM-time	0.75	0.44	0.87	0.60	0.38	0.33	0.67	0.46
DEC	MLM	0.78	0.51	0.87	0.58	0.33	0.32	0.66	0.47
DEC	MLM-time	0.77	0.47	**0.89**	0.61	0.32	0.31	0.66	0.46
$K = 50$ *clusters*									
k-means	Sensor hist.	0.81	0.61	0.88	0.63	0.35	0.34	0.68	0.53
k-means	Sensor+ToD hist.	0.79	0.58	0.87	0.63	**0.81**	**0.61**	0.82	**0.61**
k-means	MLM	0.81	0.61	**0.89**	**0.65**	0.37	0.36	0.69	0.54
k-means	MLM-time	0.81	0.60	**0.88**	**0.65**	0.37	0.34	0.69	0.53
k-means	MLM + ToD	0.81	0.55	0.88	0.60	**0.81**	0.58	**0.83**	0.58
k-means	MLM-time + ToD	**0.82**	0.61	0.86	0.61	**0.81**	0.58	**0.83**	0.60
SCAN	MLM	0.81	0.61	0.88	0.63	0.39	0.36	0.69	0.53
SCAN	MLM-time	0.81	**0.62**	**0.88**	**0.64**	0.37	0.36	0.69	0.54
DEC	MLM	0.79	0.57	0.87	0.58	0.34	0.33	0.67	0.49
DEC	MLM-time	0.80	0.57	0.88	0.61	0.34	0.31	0.67	0.50
$K = 100$ *clusters*									
k-means	Sensor hist.	0.81	0.62	**0.89**	0.65	0.35	0.35	0.68	0.54
k-means	Sensor+ToD hist.	0.81	0.63	0.87	0.63	0.81	**0.65**	0.83	0.64
k-means	MLM	0.81	**0.63**	**0.89**	0.65	0.38	0.38	0.69	0.55
k-means	MLM-time	0.81	0.62	0.88	0.64	0.37	0.36	0.69	0.54
k-means	MLM + ToD	0.82	0.63	**0.89**	**0.66**	**0.84**	**0.65**	**0.85**	**0.65**
k-means	MLM-time + ToD	**0.82**	**0.63**	0.88	0.64	0.82	0.63	0.84	0.64
SCAN	MLM	0.80	0.59	**0.89**	0.64	0.38	0.38	0.69	0.54
SCAN	MLM-time	0.81	0.60	**0.89**	0.64	0.38	0.38	0.69	0.54
DEC	MLM	0.80	0.58	0.87	0.59	0.34	0.33	0.67	0.50
DEC	MLM-time	0.79	0.56	**0.89**	0.63	0.36	0.34	0.68	0.51

Table 8: Clustering, L1, all classes; $m = all$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ *clusters*									
k-means	Sensor hist.	0.79	0.42	0.87	0.46	0.32	0.33	0.66	**0.40**
k-means	Sensor+ToD hist.	0.70	0.28	0.80	0.27	0.77	0.46	0.76	0.34
k-means	MLM	0.79	**0.43**	**0.88**	**0.49**	0.34	0.29	0.67	**0.41**
k-means	MLM-time	**0.80**	**0.43**	**0.88**	0.46	0.35	0.32	0.67	**0.40**
k-means	MLM + ToD	0.72	0.27	0.80	0.26	0.78	**0.47**	**0.77**	0.34
k-means	MLM-time + ToD	0.71	0.27	0.78	0.22	**0.79**	0.47	0.76	0.32
SCAN	MLM	0.76	0.35	0.84	0.37	0.36	0.30	0.65	0.34
SCAN	MLM-time	0.75	0.32	0.87	0.47	0.38	0.30	0.66	0.36
DEC	MLM	0.78	0.37	0.87	0.40	0.33	0.29	0.66	0.35
DEC	MLM-time	0.77	0.34	**0.88**	0.43	0.32	0.28	0.66	0.35
$K = 50$ *clusters*									
k-means	Sensor hist.	0.80	**0.50**	0.88	0.51	0.35	0.33	0.68	0.45
k-means	Sensor+ToD hist.	0.79	0.43	0.87	0.44	**0.80**	**0.56**	0.82	0.47
k-means	MLM	0.80	**0.50**	**0.89**	0.51	0.37	0.37	0.69	0.46
k-means	MLM-time	0.81	0.49	**0.88**	**0.52**	0.37	0.35	0.69	0.46
k-means	MLM + ToD	0.80	0.41	0.88	0.42	**0.81**	0.55	**0.83**	0.46
k-means	MLM-time + ToD	**0.81**	0.46	0.86	0.47	**0.81**	0.52	**0.83**	**0.48**
SCAN	MLM	0.80	0.45	0.88	**0.53**	0.39	0.32	0.69	0.43
SCAN	MLM-time	0.80	0.46	**0.88**	**0.53**	0.37	0.36	0.69	0.45
DEC	MLM	0.79	0.42	0.87	0.46	0.34	0.30	0.67	0.39
DEC	MLM-time	0.79	0.45	0.88	0.48	0.34	0.28	0.67	0.40
$K = 100$ *clusters*									
k-means	Sensor hist.	0.81	0.52	**0.88**	0.53	0.35	0.36	0.68	0.47
k-means	Sensor+ToD hist.	0.81	0.47	0.87	0.48	0.81	**0.62**	0.83	**0.52**
k-means	MLM	0.81	**0.52**	**0.89**	0.53	0.38	0.37	0.69	0.47
k-means	MLM-time	0.81	0.51	0.88	0.51	0.37	0.37	0.69	0.46
k-means	MLM + ToD	0.81	0.46	**0.89**	0.46	**0.83**	0.61	**0.84**	0.51
k-means	MLM-time + ToD	**0.82**	0.47	0.88	0.48	0.82	0.58	0.84	0.51
SCAN	MLM	0.80	0.48	**0.89**	**0.53**	0.37	0.34	0.69	0.45
SCAN	MLM-time	0.80	0.49	**0.88**	**0.53**	0.39	0.38	0.69	0.47
DEC	MLM	0.79	0.46	0.87	0.48	0.34	0.30	0.67	0.41
DEC	MLM-time	0.79	0.46	**0.89**	0.50	0.36	0.31	0.68	0.42

Table 9: Clustering, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS; $m = all$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ *clusters*									
k-means	Sensor hist.	0.79	0.45	0.88	0.53	0.76	0.33	0.81	0.44
k-means	Sensor+ToD hist.	0.72	0.31	0.80	0.31	0.84	0.40	0.79	0.34
k-means	MLM	0.80	0.45	**0.89**	**0.55**	0.76	0.35	0.82	**0.45**
k-means	MLM-time	**0.81**	**0.47**	0.89	0.50	0.78	0.33	**0.82**	0.43
k-means	MLM + ToD	0.72	0.31	0.80	0.31	**0.86**	**0.43**	0.79	0.35
k-means	MLM-time + ToD	0.72	0.31	0.78	0.28	**0.86**	0.41	0.79	0.33
SCAN	MLM	0.77	0.34	0.85	0.42	0.79	0.35	0.80	0.37
SCAN	MLM-time	0.75	0.30	0.88	**0.55**	0.77	0.32	0.80	0.39
DEC	MLM	0.78	0.35	0.88	0.46	0.77	0.33	0.81	0.38
DEC	MLM-time	0.77	0.34	**0.89**	0.47	0.75	0.29	0.81	0.37
$K = 50$ *clusters*									
k-means	Sensor hist.	0.81	0.47	0.89	0.58	0.78	0.34	0.82	0.47
k-means	Sensor+ToD hist.	0.79	0.47	0.88	0.49	0.87	0.43	0.85	0.46
k-means	MLM	0.81	0.47	**0.89**	0.58	0.79	0.42	0.83	**0.49**
k-means	MLM-time	0.81	0.48	**0.89**	0.59	0.80	0.39	0.83	0.49
k-means	MLM + ToD	0.81	0.44	0.88	0.48	**0.87**	**0.46**	**0.85**	0.46
k-means	MLM-time + ToD	**0.82**	**0.50**	0.86	0.54	**0.88**	0.44	**0.85**	**0.49**
SCAN	MLM	0.81	0.48	0.88	**0.60**	0.81	0.38	0.83	0.48
SCAN	MLM-time	0.80	0.48	**0.89**	0.59	0.80	0.36	0.83	0.48
DEC	MLM	0.80	0.44	0.88	0.53	0.80	0.37	0.83	0.44
DEC	MLM-time	0.80	0.43	0.89	0.53	0.78	0.34	0.82	0.43
$K = 100$ *clusters*									
k-means	Sensor hist.	0.81	0.49	**0.89**	0.59	0.79	0.35	0.83	0.48
k-means	Sensor+ToD hist.	0.82	0.50	0.88	0.52	0.87	**0.49**	0.86	**0.51**
k-means	MLM	0.82	0.50	**0.89**	0.59	0.81	0.43	0.84	**0.51**
k-means	MLM-time	0.81	0.50	0.89	0.57	0.81	0.45	0.84	**0.51**
k-means	MLM + ToD	**0.82**	0.50	**0.89**	0.50	**0.89**	0.47	**0.87**	0.49
k-means	MLM-time + ToD	**0.82**	**0.51**	0.88	0.54	0.88	0.47	0.86	**0.50**
SCAN	MLM	0.80	0.45	**0.89**	0.60	0.82	0.40	0.84	0.48
SCAN	MLM-time	0.81	0.47	**0.89**	**0.60**	0.82	0.45	0.84	**0.51**
DEC	MLM	0.80	0.45	0.87	0.54	0.79	0.36	0.82	0.45
DEC	MLM-time	0.80	0.43	**0.89**	0.55	0.80	0.38	0.83	0.45

Linear probes (logistic regression on frozen representations)

Table 10: Linear probes, L1 (per-house activities, resident id removed), rare classes masked. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
100 random labelled windows									
Linear probe	Sensor hist.	0.75	0.47	0.86	0.60	0.34	0.26	0.65	0.44
Linear probe	Sensor+ToD hist.	**0.76**	**0.51**	0.86	0.60	**0.77**	**0.52**	**0.80**	**0.54**
Linear probe	MLM	0.74	0.46	**0.87**	**0.61**	0.35	0.28	0.65	0.45
Linear probe	MLM-time	**0.76**	0.49	**0.87**	0.60	0.48	0.36	0.70	0.48
Linear probe	MLM + ToD	0.75	0.48	**0.87**	**0.61**	0.68	0.44	0.77	0.51
Linear probe	MLM-time + ToD	**0.76**	**0.50**	**0.87**	0.60	0.65	0.45	0.76	0.52
500 random labelled windows									
Linear probe	Sensor hist.	0.80	0.59	0.87	0.62	0.36	0.35	0.68	0.52
Linear probe	Sensor+ToD hist.	**0.80**	**0.61**	**0.87**	0.63	**0.82**	**0.64**	**0.83**	**0.63**
Linear probe	MLM	0.78	0.56	**0.88**	0.63	0.38	0.37	0.68	0.52
Linear probe	MLM-time	0.80	0.59	**0.88**	**0.64**	0.70	0.55	0.79	0.60
Linear probe	MLM + ToD	**0.80**	0.59	**0.88**	0.63	0.81	0.64	**0.83**	0.62
Linear probe	MLM-time + ToD	**0.80**	**0.60**	**0.88**	**0.64**	0.81	**0.64**	**0.83**	**0.63**
all labelled windows (max 20k)									
Linear probe	Sensor hist.	0.82	0.63	0.90	0.66	0.39	0.39	0.70	0.56
Linear probe	Sensor+ToD hist.	**0.84**	**0.67**	**0.90**	0.67	0.84	0.69	**0.86**	0.68
Linear probe	MLM	0.83	0.64	**0.90**	0.67	0.43	0.44	0.72	0.59
Linear probe	MLM-time	0.83	0.64	**0.90**	**0.69**	0.84	0.70	0.86	0.68
Linear probe	MLM + ToD	**0.84**	**0.67**	**0.90**	0.67	**0.85**	**0.72**	**0.86**	**0.69**
Linear probe	MLM-time + ToD	0.83	0.64	**0.90**	**0.69**	**0.85**	0.71	**0.86**	0.68

Table 11: Linear probes, L1, all classes. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
100 random labelled windows									
Linear probe	Sensor hist.	0.74	0.35	0.86	0.42	0.33	0.24	0.65	0.34
Linear probe	Sensor+ToD hist.	**0.75**	**0.37**	0.86	0.42	**0.77**	**0.48**	**0.79**	**0.42**
Linear probe	MLM	0.73	0.34	**0.87**	**0.43**	0.35	0.25	0.65	0.34
Linear probe	MLM-time	**0.75**	0.37	**0.87**	0.42	0.47	0.32	0.70	0.37
Linear probe	MLM + ToD	0.74	0.36	**0.87**	**0.43**	0.68	0.40	0.76	0.40
Linear probe	MLM-time + ToD	**0.76**	**0.37**	**0.87**	0.42	0.65	0.41	0.76	0.40
500 random labelled windows									
Linear probe	Sensor hist.	0.79	0.47	0.87	0.47	0.36	0.36	0.67	0.43
Linear probe	Sensor+ToD hist.	**0.80**	0.49	**0.87**	0.48	**0.82**	0.61	**0.83**	0.52
Linear probe	MLM	0.78	0.45	**0.88**	0.47	0.38	0.37	0.68	0.43
Linear probe	MLM-time	0.79	0.47	**0.88**	**0.48**	0.69	0.54	0.79	0.50
Linear probe	MLM + ToD	**0.80**	0.48	**0.88**	0.48	0.81	0.61	**0.83**	0.52
Linear probe	MLM-time + ToD	**0.80**	**0.49**	**0.88**	**0.48**	0.81	**0.62**	**0.83**	**0.53**
all labelled windows (max 20k)									
Linear probe	Sensor hist.	0.82	0.54	0.89	0.67	0.39	0.41	0.70	0.54
Linear probe	Sensor+ToD hist.	0.83	0.53	**0.90**	0.69	0.84	0.68	**0.86**	0.63
Linear probe	MLM	0.83	**0.57**	**0.90**	0.69	0.43	0.46	0.72	0.58
Linear probe	MLM-time	0.82	0.56	**0.90**	0.73	0.84	0.70	0.86	**0.66**
Linear probe	MLM + ToD	**0.84**	**0.57**	**0.90**	0.71	**0.85**	**0.71**	**0.86**	**0.66**
Linear probe	MLM-time + ToD	0.82	0.55	**0.90**	**0.73**	**0.85**	0.71	**0.86**	**0.67**

Table 12: Linear probes, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
100 random labelled windows									
Linear probe	Sensor hist.	0.76	0.36	0.87	0.47	0.73	0.31	0.78	0.38
Linear probe	Sensor+ToD hist.	**0.77**	**0.40**	0.87	0.47	**0.83**	**0.43**	**0.82**	**0.43**
Linear probe	MLM	0.74	0.36	**0.88**	**0.49**	0.75	0.33	0.79	0.39
Linear probe	MLM-time	0.76	0.38	**0.88**	0.48	0.79	0.37	0.81	0.41
Linear probe	MLM + ToD	0.75	0.38	**0.88**	**0.49**	0.79	0.38	0.81	0.41
Linear probe	MLM-time + ToD	0.76	0.39	**0.88**	0.48	0.81	0.40	**0.82**	0.42
500 random labelled windows									
Linear probe	Sensor hist.	0.80	0.46	0.88	0.52	0.80	0.42	0.83	0.47
Linear probe	Sensor+ToD hist.	**0.81**	**0.50**	**0.88**	**0.53**	**0.88**	**0.52**	**0.86**	**0.52**
Linear probe	MLM	0.79	0.44	**0.88**	0.52	0.80	0.44	0.82	0.47
Linear probe	MLM-time	0.79	0.46	**0.88**	0.52	0.85	0.48	0.84	0.49
Linear probe	MLM + ToD	0.80	0.47	**0.88**	**0.53**	0.87	0.52	0.85	0.51
Linear probe	MLM-time + ToD	0.80	0.48	**0.88**	0.52	0.87	0.52	0.85	0.51
all labelled windows (max 20k)									
Linear probe	Sensor hist.	0.82	0.49	0.90	0.77	0.82	0.46	0.85	0.57
Linear probe	Sensor+ToD hist.	**0.84**	0.56	0.90	0.78	**0.91**	0.59	**0.88**	0.65
Linear probe	MLM	0.83	0.55	**0.91**	0.79	0.83	0.52	0.86	0.62
Linear probe	MLM-time	0.83	0.57	**0.91**	**0.83**	0.90	0.65	0.88	**0.68**
Linear probe	MLM + ToD	**0.84**	**0.59**	**0.91**	0.79	**0.91**	**0.65**	**0.89**	0.68
Linear probe	MLM-time + ToD	0.83	0.57	**0.91**	**0.83**	**0.91**	**0.66**	**0.88**	**0.68**

Linear probe vs clustering at matched label budgets

Table 13: Probe vs clustering at matched label budgets, L1 (per-house activities, resident id removed), rare classes masked. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	K20, m5		100 labels		K100, m5		500 labels		K100, m all		all labels	
		W	M	W	M	W	M	W	M	W	M	W	M
three-house average; clustering columns = k-means / SCAN / DEC, probe columns = same representation													
k-means / probe	Sensor hist.	0.61	0.45	0.65	0.44	0.67	0.51	0.68	0.52	0.68	0.54	0.70	0.56
k-means / probe	Sensor+ToD hist.	0.75	0.41	**0.80**	**0.54**	0.83	0.60	**0.83**	**0.63**	0.83	0.64	**0.86**	0.68
k-means / probe	MLM	0.65	**0.48**	0.65	0.45	0.69	0.53	0.68	0.52	0.69	0.55	0.72	0.59
k-means / probe	MLM-time	0.63	0.45	0.70	0.48	0.67	0.53	0.79	0.60	0.69	0.54	0.86	0.68
k-means / probe	MLM + ToD	0.77	0.40	0.77	0.51	**0.84**	**0.61**	**0.83**	0.62	**0.85**	**0.65**	**0.86**	**0.69**
k-means / probe	MLM-time + ToD	**0.77**	0.44	0.76	0.52	**0.83**	0.60	**0.83**	**0.63**	0.84	0.64	**0.86**	0.68
SCAN	MLM	0.66	0.41	–	–	0.68	0.52	–	–	0.69	0.54	–	–
DEC	MLM	0.63	0.45	–	–	0.66	0.48	–	–	0.67	0.50	–	–

Table 14: Probe vs clustering at matched label budgets, L1, all classes. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	K20, m5		100 labels		K100, m5		500 labels		K100, m all		all labels	
		W	M	W	M	W	M	W	M	W	M	W	M
three-house average; clustering columns = k-means / SCAN / DEC, probe columns = same representation													
k-means / probe	Sensor hist.	0.61	**0.36**	0.65	0.34	0.67	0.42	0.67	0.43	0.68	0.47	0.70	0.54
k-means / probe	Sensor+ToD hist.	0.75	0.32	**0.79**	**0.42**	0.82	**0.49**	**0.83**	0.52	0.83	**0.52**	**0.86**	0.63
k-means / probe	MLM	0.65	**0.36**	0.65	0.34	0.69	0.44	0.68	0.43	0.69	0.47	0.72	0.58
k-means / probe	MLM-time	0.63	0.34	0.70	0.37	0.67	0.40	0.79	0.50	0.69	0.46	0.86	**0.66**
k-means / probe	MLM + ToD	0.76	0.32	0.76	0.40	**0.83**	0.48	**0.83**	0.52	**0.84**	0.51	**0.86**	**0.66**
k-means / probe	MLM-time + ToD	**0.77**	0.35	0.76	0.40	**0.83**	0.48	**0.83**	**0.53**	0.84	0.51	**0.86**	**0.67**
SCAN	MLM	0.65	0.31	–	–	0.68	0.45	–	–	0.69	0.45	–	–
DEC	MLM	0.62	0.34	–	–	0.66	0.38	–	–	0.67	0.41	–	–

Table 15: Probe vs clustering at matched label budgets, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	K20, m5		100 labels		K100, m5		500 labels		K100, m all		all labels	
		W	M	W	M	W	M	W	M	W	M	W	M
three-house average; clustering columns = k-means / SCAN / DEC, probe columns = same representation													
k-means / probe	Sensor hist.	0.77	**0.39**	0.78	0.38	0.81	0.42	0.83	0.47	0.83	0.48	0.85	0.57
k-means / probe	Sensor+ToD hist.	0.78	0.33	**0.82**	**0.43**	0.85	0.47	**0.86**	**0.52**	0.86	**0.51**	**0.88**	0.65
k-means / probe	MLM	0.80	0.38	0.79	0.39	0.83	0.46	0.82	0.47	0.84	**0.51**	0.86	0.62
k-means / probe	MLM-time	0.79	0.38	0.81	0.41	0.82	0.43	0.84	0.49	0.84	**0.51**	0.88	**0.68**
k-means / probe	MLM + ToD	0.80	0.35	0.81	0.41	**0.86**	0.46	0.85	0.51	**0.87**	0.49	**0.89**	0.68
k-means / probe	MLM-time + ToD	0.79	0.35	**0.82**	0.42	0.85	0.48	0.85	0.51	0.86	**0.50**	**0.88**	**0.68**
SCAN	MLM	0.79	0.34	–	–	0.83	**0.48**	–	–	0.84	0.48	–	–
DEC	MLM	**0.81**	**0.39**	–	–	0.82	0.41	–	–	0.82	0.45	–	–